

Jin Cheng

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EDUCATION

Doctoral candidate in Computer Science, ETH Zürich, Switzerland	since 10/2023
<i>Advisor: Prof. Dr. Stelian Coros (ETH Zürich), Prof. Dr. Guanya Shi (Carnegie Mellon University)</i>	
M.Sc. in Mechanical Engineering, ETH Zürich, Switzerland	09/2020 – 07/2023
B.E. in Vehicle Engineering, Tsinghua University, China	09/2016 – 07/2020

EXPERIENCE

Scientific Assistant, ETH Zürich	since 10/2023
<i>Computational Robotics Lab (CRL), supervised by Prof. Dr. Stelian Coros</i>	
Research Intern, Max Planck Institute for Intelligent Systems	04/2023 - 09/2023
<i>Autonomous Learning Group, supervised by Prof. Dr. Georg Martius</i>	
Research Assistant, ETH Zürich	11/2022 - 03/2023
<i>Computational Robotics Lab (CRL), supervised by Prof. Dr. Stelian Coros</i>	
Semester and Master Thesis, ETH Zürich	09/2021 - 11/2022
<i>Robotic Systems Lab (RSL), supervised by Prof. Dr. Marco Hutter</i>	
Research Assistant and Bachelor Thesis, Tsinghua University	09/2019 - 05/2020
<i>State Key Laboratory of Intelligent Green Vehicle and Mobility, supervised by Prof. Dr. Qing Zhou</i>	

PUBLICATIONS

Maximum Entropy Behavior Exploration for Sim2Real Zero-Shot Reinforcement Learning (RLC 2026)	
<i>Jiajun Hu, Núria Armengol Urpí, Jin Cheng, Stelian Coros</i>	05/2026
CAIMAN: Causal Action Influence Detection for Sample-Efficient Loco-Manipulation (ICRA 2026)	
<i>Yuanchen Yuan, Jin Cheng, Núria Armengol Urpí, Stelian Coros</i>	02/2026
Benchmarking Whole-Body Motion Tracking Policies via Just Dance: Using a Commercial Console Game for Embodied-AI Evaluation (CVPR Findings 2026)	
<i>Jeonghwan Kim, Wontaek Kim, Yidan Lu, Jin Cheng, Fatemeh Zargarbashi, Zicheng Zeng, Zekun Qi, Zhiyang Dou, Nitish Sontakke, Donghoon Baek, Li Yi, Sehoon Ha, Tianyu Li</i>	02/2026
Whole-Body Inverse Dynamics MPC for Legged Loco-Manipulation (RA-L)	
<i>Lukas Molnar, Jin Cheng, Gabriele Fadini, Donghoon Kang, Fatemeh Zargarbashi, Stelian Coros</i>	11/2025
RAMBO: RL-Augmented Model-Based Whole-Body Control for Loco-Manipulation (RA-L)	
<i>Jin Cheng, Donghoon Kang, Gabriele Fadini, Guanya Shi, Stelian Coros</i>	07/2025
Spatio-Temporal Motion Retargeting for Quadruped Robots (T-RO)	
<i>Taerim Yoon, Donghoon Kang, Seungmin Kim, Jin Cheng, Minsung Ahn, Stelian Coros, Sungjoon Choi</i>	07/2025
SATA: Safe and Adaptive Torque-Based Locomotion Policies Inspired by Animal Learning (RSS 2025)	
<i>Peizhuo Li, Hongyi Li, Ge Sun, Jin Cheng, Xinrong Yang, Guillaume Bellegarda, Milad Shafiee, Yuhong Cao, Auke Ijspeert, Guillaume Sartoretti</i>	02/2025
DARE: Diffusion Policy for Autonomous Robot Exploration (ICRA 2025)	
<i>Yuhong Cao, Jeric Lew, Jingsong Liang, Jin Cheng, Guillaume Sartoretti</i>	02/2025

RobotKeyframing: Learning Locomotion with High-Level Objectives via Mixture of Dense and Sparse Rewards (CoRL 2024) <i>Fatemeh Zargarbashi, Jin Cheng, Dongho Kang, Robert Sumner, and Stelian Coros</i>	09/2024
Offline Diversity Maximization Under Imitation Constraints (RLC 2024) <i>Marin Vlastelica, Jin Cheng, Georg Martius, Pavel Kolev</i>	05/2024
Learning Diverse Skills for Local Navigation under Multi-constraint Optimality (ICRA 2024) <i>Jin Cheng, Marin Vlastelica, Pavel Kolev, Chenhao Li, Georg Martius</i>	02/2024
RL + Model-based Control: Using On-demand Optimal Control to Learn Versatile Legged Locomotion (RA-L) <i>Dongho Kang, Jin Cheng, Miguel Zamora, Fatemeh Zargarbashi, Stelian Coros</i>	09/2023
Haptic Teleoperation of High-dimensional Robotic Systems Using a Feedback MPC Framework (IROS 2022) <i>Jin Cheng, Firas Abi-Farraj, Farbod Farshidian, Marco Hutter</i>	10/2022

PREPRINTS

Teaching Robots Like Dogs: Learning Agile Navigation from Luring, Gesture, and Speech (Under Review) <i>Taerim Yoon, Dongho Kang, Jin Cheng, Fatemeh Zargarbashi, Yijiang Huang, Minsung Ahn, Stelian Coros, Sungjoon Choi</i>	01/2026
APEX: Action Priors Enable Efficient Exploration for Robust Motion Tracking on Legged Robots (Under Review) <i>Shivam Sood, Laukik Nakhwa, Ge Sun, Yuhong Cao, Jin Cheng, Fatemeh Zargarbashi, Taerim Yoon, Sungjoon Choi, Stelian Coros, Guillaume Sartoretti</i>	11/2025
Learning Steerable Imitation Controllers from Unstructured Animal Motions (Under Review) <i>Dongho Kang, Jin Cheng, Fatemeh Zargarbashi, Taerim Yoon, Sungjoon Choi, Stelian Coros</i>	07/2025
TARC: Time-Adaptive Robotic Control (Under Review) <i>Arnav Sukhija, Lenart Treven, Jin Cheng, Florian Dörfler, Stelian Coros, Andreas Krause</i>	09/2025
Learning More With Less: Sample Efficient Dynamics Learning and Model-Based RL for Loco-Manipulation (arXiv) <i>Benjamin Hoffman, Jin Cheng, Chenhao Li, Stelian Coros</i>	02/2025
MetaLoco: Universal Quadrupedal Locomotion with Meta-Reinforcement Learning and Motion Imitation (arXiv) <i>Fatemeh Zargarbashi, Fabrizio Di Giuro, Jin Cheng, Dongho Kang, Bhavya Sukhija, Stelian Coros</i>	09/2024

INVITED TALKS

Intelligent Motion Lab (IMO) @ Seoul National University (SNU)	10/2025
Frontier AI & Robotics (FAR) @ Amazon	05/2025
Robotics and AI Institute (RAI) @ Zurich	04/2025
Biomimetic Robotics Laboratory @ Massachusetts Institute of Technology (MIT)	11/2023

TEACHING

Computational Models of Motion (263-5807-00) <i>Computational Robotics Lab (CRL) at ETH Zürich, lectured by S. Coros</i>	02/2026 - 07/2026
Probabilistic Artificial Intelligence (263-5210-00) <i>Learning & Adaptive Systems Group (LAS) at ETH Zürich, lectured by A. Krause</i>	08/2025 - 02/2026
Computational Models of Motion (263-5807-00) <i>Computational Robotics Lab (CRL) at ETH Zürich, lectured by S. Coros</i>	02/2025 - 07/2025

Probabilistic Artificial Intelligence (263-5210-00) <i>Learning & Adaptive Systems Group (LAS) at ETH Zürich, lectured by A. Krause</i>	08/2024 - 02/2025
Introduction to Machine Learning (252-0220-00) <i>Learning & Adaptive Systems Group (LAS) at ETH Zürich, lectured by F. Perez Cruz, F. Yang</i>	02/2024 - 08/2024
Dynamic Programming and Optimal Control (151-0563-01) <i>Institute for Dynamic Systems and Control (IDSC) at ETH Zürich, lectured by R. D'Andrea</i>	08/2022 - 02/2023
Recursive Estimation (151-0566-00) <i>Institute for Dynamic Systems and Control (IDSC) at ETH Zürich, lectured by R. D'Andrea</i>	02/2022 - 08/2022
Dynamic Programming and Optimal Control (151-0563-01) <i>Institute for Dynamic Systems and Control (IDSC) at ETH Zürich, lectured by R. D'Andrea</i>	08/2021 - 02/2022

SKILLS AND SERVICES

Language: Chinese (Native), English (C1), German (B2)
Programming: Python, C++, Git, Docker, L^AT_EX, PyTorch, ROS
Journal Reviewer: T-RO, RA-L
Conference Reviewer: CoRL, RSS, ICRA, IROS, RLC, SIGGRAPH

SCHOLARSHIPS AND AWARDS

Outstanding Teaching Assistant Award <i>ETH Zürich</i>	03/2022 Zürich, Switzerland
Friends of Tsinghua Scholarship – German Scholarship <i>Tsinghua University</i>	10/2019 Beijing, China
Academic Excellence Scholarship <i>Tsinghua University</i>	10/2018, 10/2019 Beijing, China
Volunteer Public Service Scholarship <i>Tsinghua University</i>	10/2018 Beijing, China
Integrated Excellence Scholarship <i>Tsinghua University</i>	10/2017 Beijing, China

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